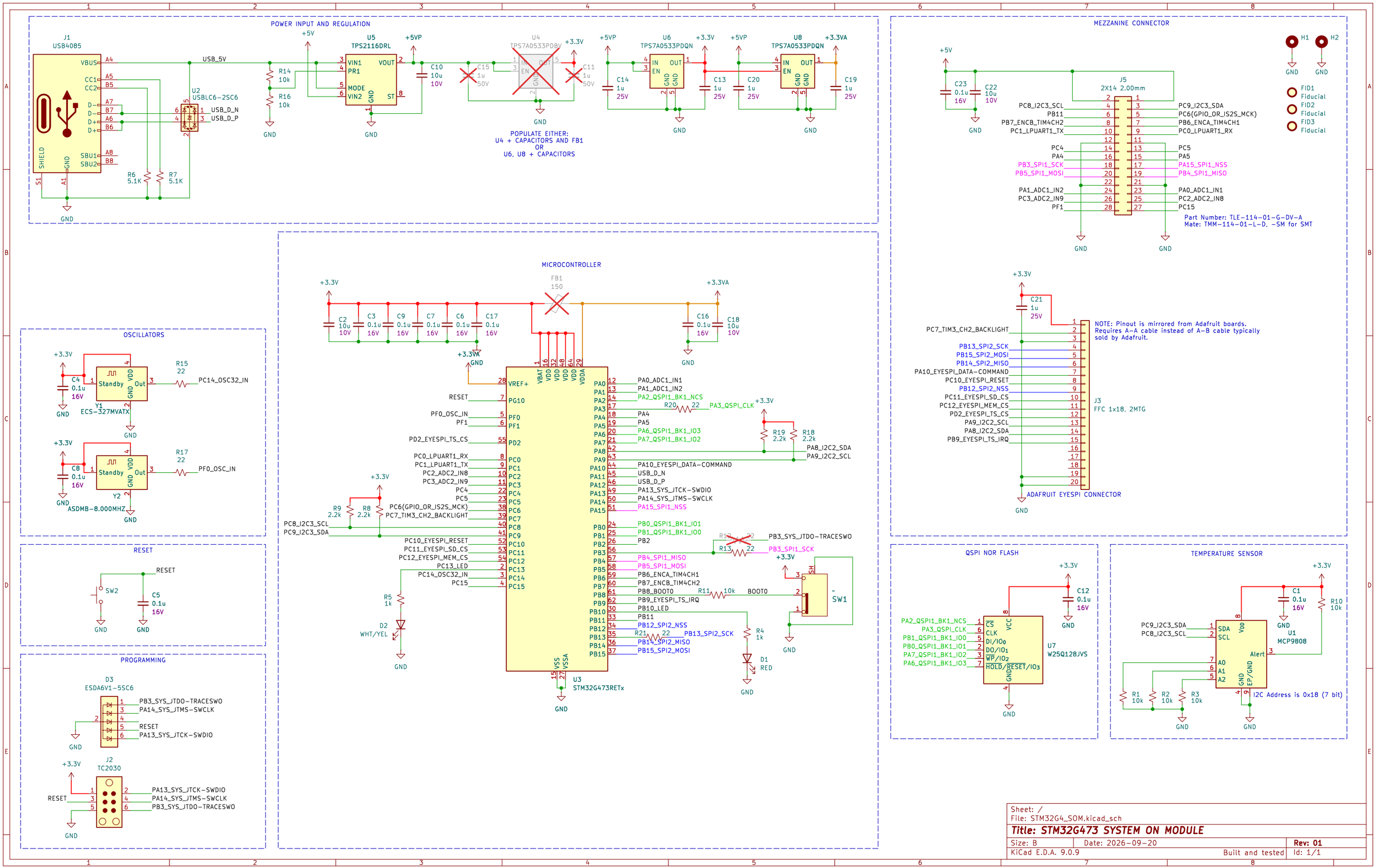

POWER INPUT AND REGULATION
J1
USB4085
VBUS
CC1
CC2
D−
D−
D+
D+
SHIELD
GND
SBU1
SBU2
A4
A5
B5
A7
B7
A6
B6
A8
B8
S1
A1
USB_5V
U2
USBLC6−2SC6
USB_D_N
USB_D_P
R6
5.1K
R7
5.1K
GND
+5V
R14
10k
R16
10k
U5
TPS2116DRL
VIN1
PR1
MODE
VIN2
VOUT
GND
ST
+5VP
C10
10u
10V
C15
1u
50V
U4
TPS7A0533PDBV
IN
EN
OUT
GND
+3.3V
C11
1u
50V
POPULATE EITHER:
U4 + CAPACITORS AND FB1
OR
U6, U8 + CAPACITORS
C14
1u
25V
U6
TPS7A0533PDQN
C13
1u
25V
C20
1u
25V
U8
TPS7A0533PDQN
C19
1u
25V
+3.3VA
MEZZANINE CONNECTOR
H1
H2
FID1
Fiducial
FID2
Fiducial
FID3
Fiducial
C23
0.1u
16V
C22
10u
10V
J5
2X14 2.00mm
PC8_I2C3_SCL
PB11
PB7_ENCB_TIM4CH2
PC1_LPUART1_TX
PC4
PA4
PB3_SPI1_SCK
PB5_SPI1_MOSI
PA1_ADC1_IN2
PC3_ADC2_IN9
PF1
PC9_I2C3_SDA
PC6(GPIO_OR_IS2S_MCK)
PB6_ENCA_TIM4CH1
PC0_LPUART1_RX
PC5
PA5
PA15_SPI1_NSS
PB4_SPI1_MISO
PA0_ADC1_IN1
PC2_ADC2_IN8
PC15
Part Number: TLE−114−01−G−DV−A
Mate: TMM−114−01−L−D, −SM for SMT
OSCILLATORS
C4
0.1u
16V
Standby
VDD
GND
Out
Y1
ECS−327MVATX
R15
22
PC14_OSC32_IN
C8
0.1u
16V
Y2
ASDMB−8.000MHZ
R17
22
PF0_OSC_IN
RESET
SW2
C5
0.1u
16V
PROGRAMMING
D3
ESDA6V1−5SC6
PB3_SYS_JTDO−TRACESWO
PA14_SYS_JTMS−SWCLK
RESET
PA13_SYS_JTCK−SWDIO
J2
TC2030
MICROCONTROLLER
FB1
150
C2
10u
10V
C3
0.1u
16V
C9
0.1u
16V
C7
0.1u
16V
C6
0.1u
16V
C17
0.1u
16V
C16
0.1u
16V
C18
10u
10V
VBAT
VDD
VDD
VDD
VDD
VDDA
VREF+
PG10
PF0
PF1
PD2
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
VSS
VSSA
U3
STM32G473RETx
PA0_ADC1_IN1
PA1_ADC1_IN2
PA2_QSPI1_BK1_NCS
PA3_QSPI1_CLK
R20
22
PA4
PA5
PA6_QSPI1_BK1_IO3
PA7_QSPI1_BK1_IO2
PA8_I2C2_SDA
PA9_I2C2_SCL
PA10_EYESPI_DATA−COMMAND
USB_D_N
USB_D_P
PA13_SYS_JTCK−SWDIO
PA14_SYS_JTMS−SWCLK
PA15_SPI1_NSS
PB0_QSPI1_BK1_IO1
PB1_QSPI1_BK1_IO0
PB2
PB3_SYS_JTDO−TRACESWO
PB4_SPI1_MISO
PB5_SPI1_MOSI
PB6_ENCA_TIM4CH1
PB7_ENCB_TIM4CH2
PB8_BOOT0
PB9_EYESPI_TS_IRQ
PB10_LED
PB11
PB12_SPI2_NSS
PB13_SPI2_SCK
PB14_SPI2_MISO
PB15_SPI2_MOSI
PD2_EYESPI_TS_CS
PC0_LPUART1_RX
PC1_LPUART1_TX
PC2_ADC2_IN8
PC3_ADC2_IN9
PC4
PC5
PC6(GPIO_OR_IS2S_MCK)
PC7_TIM3_CH2_BACKLIGHT
PC8_I2C3_SCL
PC9_I2C3_SDA
PC10_EYESPI_RESET
PC11_EYESPI_SD_CS
PC12_EYESPI_MEM_CS
PC13_LED
PC14_OSC32_IN
PC15
R19
2.2k
R18
2.2k
R9
2.2k
R8
2.2k
R5
1k
D2
WHT/YEL
R12
22
R13
22
PB3_SPI1_SCK
R11
10k
BOOT0
SW1
R21
22
PB13_SPI2_SCK
R4
1k
D1
RED
NOTE: Pinout is mirrored from Adafruit boards.
Requires A−A cable instead of A−B cable typically
sold by Adafruit.
C21
1u
25V
PC7_TIM3_CH2_BACKLIGHT
PB13_SPI2_SCK
PB15_SPI2_MOSI
PB14_SPI2_MISO
PA10_EYESPI_DATA−COMMAND
PC10_EYESPI_RESET
PB12_SPI2_NSS
PC11_EYESPI_SD_CS
PC12_EYESPI_MEM_CS
PD2_EYESPI_TS_CS
PA9_I2C2_SCL
PA8_I2C2_SDA
PB9_EYESPI_TS_IRQ
J3
FFC 1x18, 2MTG
ADAFRUIT EYESPI CONNECTOR
QSPI NOR FLASH
C12
0.1u
16V
PA2_QSPI1_BK1_NCS
PA3_QSPI1_CLK
PB1_QSPI1_BK1_IO0
PB0_QSPI1_BK1_IO1
PA7_QSPI1_BK1_IO2
PA6_QSPI1_BK1_IO3
CS
CLK
DI/IO0
DO/IO1
WP/IO2
HOLD/RESET/IO3
VCC
GND
U7
W25Q128JVS
TEMPERATURE SENSOR
C1
0.1u
16V
R10
10k
PC9_I2C3_SDA
PC8_I2C3_SCL
SDA
SCL
Vdd
Alert
A0
A1
A2
GND
EP/GND
U1
MCP9808
R1
10k
R2
10k
R3
10k
I2C Address is 0x18 (7 bit)
Sheet: /
File: STM32G4_SOM.kicad_sch
Title: STM32G473 SYSTEM ON MODULE
Size: B
Date: 2026−09−20
Rev: 01
KiCad E.D.A. 9.0.9
Built and tested
Id: 1/1